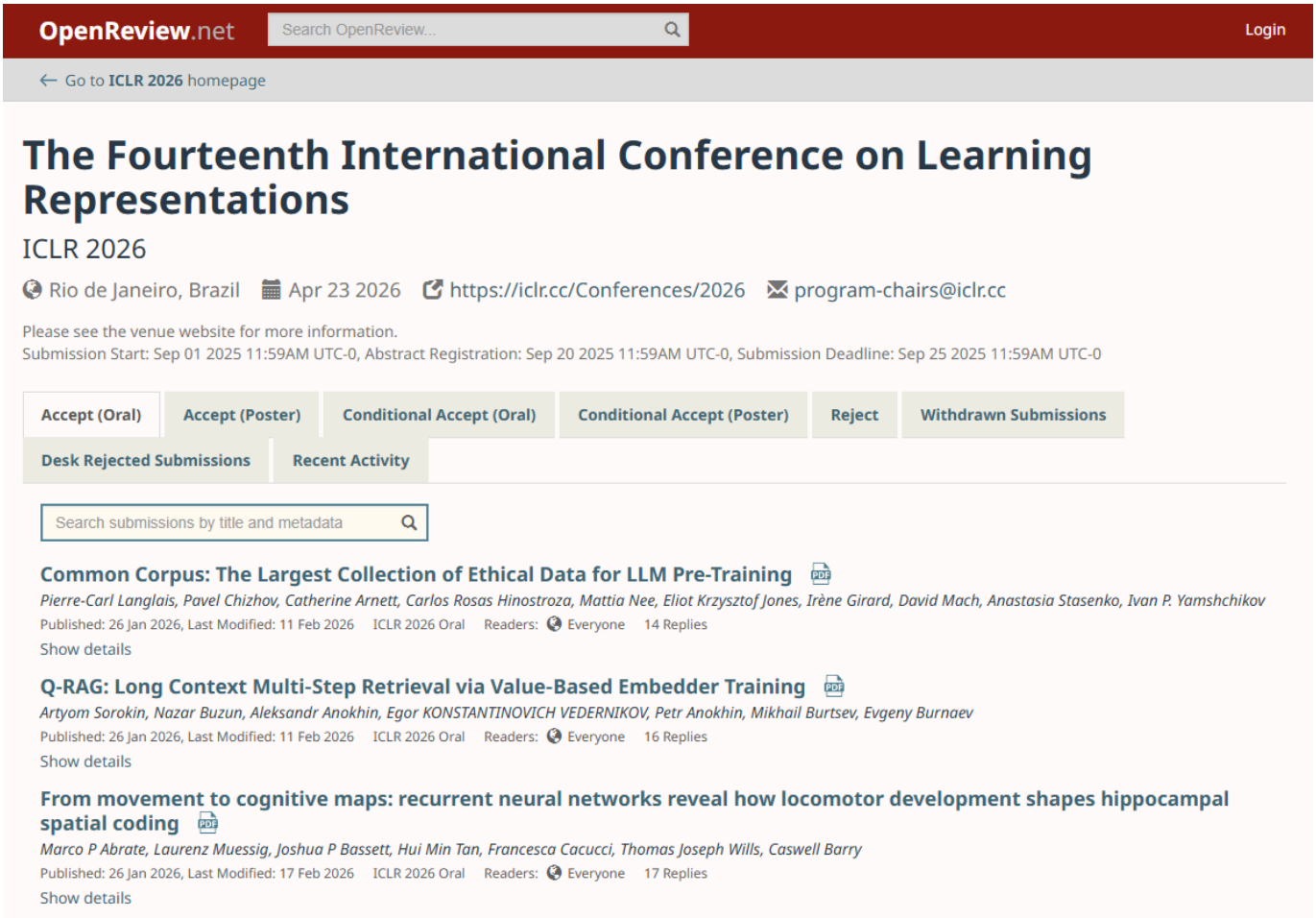
实验二

中国科大2026年春季学期"数据分析及实践"课程 - 实验二说明文档。

任务概述

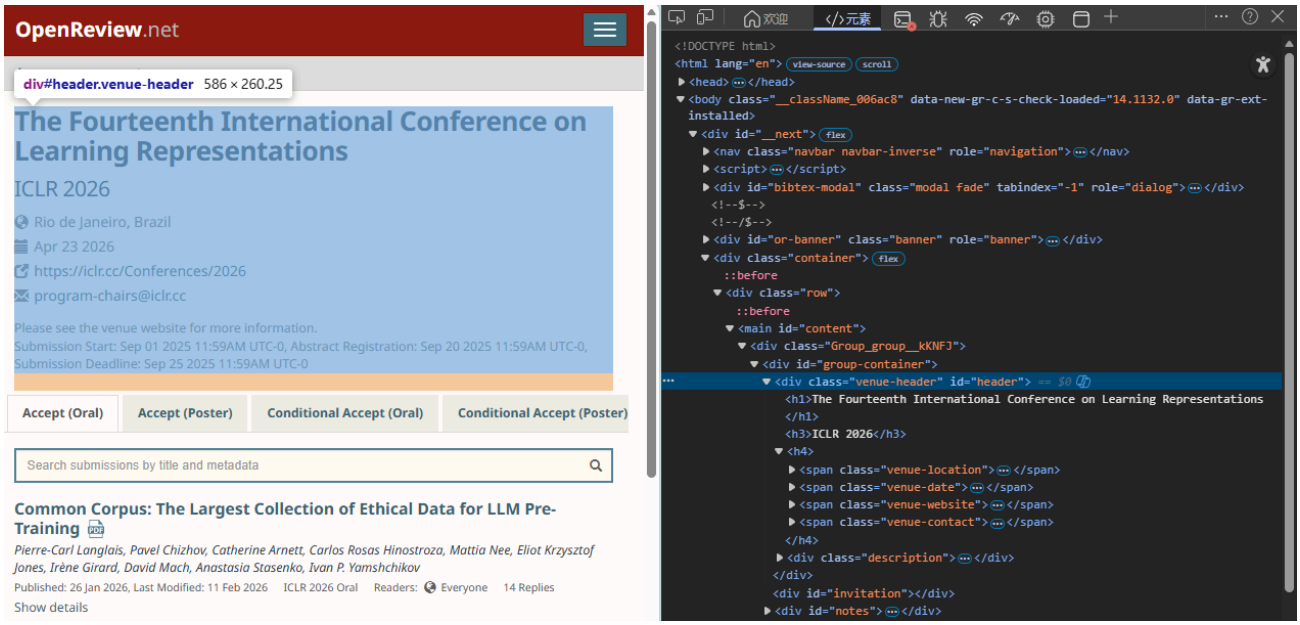
现代 Web 页面的数据加载方式多种多样，以 OpenReview 为代表的现代学术平台与传统的静态网站（如 DBLP）有着显著的区别，这对网络数据的采集与预处理提出了新的要求。 现欲通过 OpenReview 平台获取人工智能领域顶级学术会议 ICLR (International Conference on Learning Representations) 2026 的论文收录相关作信息，请你按要求编写 Python 代码实现任务列表中的内容。



任务列表

1.（30%）打开 URL 链接: https://openreview.net/group?id=ICLR.cc/2026/Conference ，进入 ICLR 2026 Conference OpenReview 主页，按要求完成以下步骤：

- 1a.（7%）使用`urllib`或者`requests`库的相关方法请求上述 ICLR 2026 主页的 URL，获取该页面的 HTML 内容，并将其以 UTF-8 编码格式写入`iclr_home.html`文件中，留待后续观察。
- 1b.（7%）使用`BeautifulSoup`库的相关方法，从该网页 HTML 内容中提取`<title>`标签内容，该内容和浏览器展示界面的哪一部分元素相对应。
- 1c.（8%）在浏览器界面按快捷键`F12`打开开发者工具，观察到该页面显示的会议时间、地点等基本信息包含在一个`<div>`标签中。请尝试使用`BeautifulSoup`库的相关方法，从该网页 HTML 内容中提取会议地点、会议时间、会议网址和联系方式等信息，观察并阐述所呈现的结果是否如你所愿。



- 1d.（8%）自动化浏览器操作工具集`Selenium`可用于 Web 应用程序的自动化测试和动态模拟。请你首先使用以下命令安装该库：

```
pip install selenium
```

随后，在脚本内执行以下代码片段：

```
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()  # 可以改为使用 Edge
driver.get(url)
```

进一步地，请你使用`driver.find_element`方法，提取会议地点、会议时间、会议网址和联系方式等信息，观察并阐述所呈现的结果是否如你所愿。

2.（28%）本任务拟使用 OpenReview 平台官方提供的 API 获取 ICLR 2026 论文基本信息数据。先设置以下变量参数：

```
api_url = "https://api2.openreview.net/notes"
params = {
  "invitation": "ICLR.cc/2026/Conference/-/Submission",
  "limit": 50,  # 单次请求返回的最大论文数
  "offset": 0  # 翻页偏移量
}
```

- 2a.（10%）请使用`requests`库相关方法，从该接口中成功获取前 1000 篇 ICLR 2026 的论文原始记录（每个批次提取`notes`中的内容），并将每一篇论文的信息追加存储到`paper_info_api_response.json`中，留待后续处理。
  - 由于 API 存在单次返回数量限制，可结合循环（不断更新`offset`参数）分页获取批量数据。
  - 为防止请求过快被服务器拦截，在每次循环获取数据后请使用`time.sleep(0.5)`暂停 0.5 秒。
  - 存储数据时可使用`json`库相关方法进行存储，并设置 2 字符缩进以保证审阅数据的美观性。
  - 最终所得数据的参考格式如下：

```
[
  {
    "content": {
      "title": {
        "value": "Improving Developer Emotion Classification via LLM-Based
Augmentation"
      },
      "authors": {
        "value": [
          "Fahmida Haque Fariha",
          "...",
        ]
      },
      ...
    },
    "id": "FPLNSx1jmL",
    ...
  },
  ...
]
```

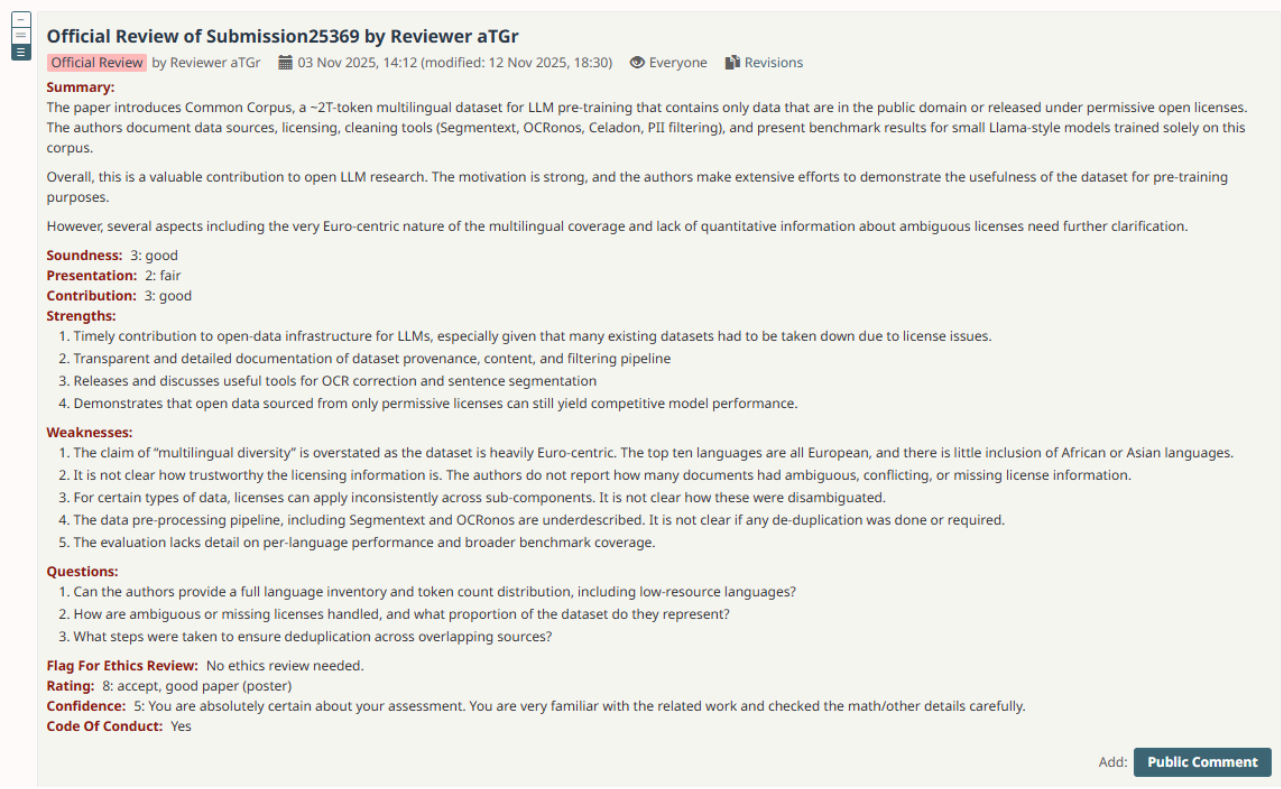
- 2b.（10%）观察单篇论文的原始字典结构可以发现，绝大部分核心信息存放在`content`键下，且内部键值又嵌套了一层`value`属性，可进行简化处理。请使用`json`包读取`paper_info_api_response.json`文件中的论文信息，从每篇论文的原始信息中提取论文唯一标识符`id`，标题`title`，作者列表`authors`，作者标识符ID`authorids`，关键词`keywords`，主要领域`primary_area`，序号`number`和收录集信息`venue`与`venueid`。将新的数据列表存储到`paper_info_cleaned.json`中，留待后续处理。
    - 最终所得数据的参考格式如下：

```
[
  {
    "id": "FPLNSx1jmL",
    "title": "Improving Developer Emotion Classification via LLM-Based
Augmentation",
    "authors": [...],
    "authorids": [...],
    "keywords": [...],
    "primary_area": ...,
    "number": 25649,
    "venue": ... ,
    "venueid": ...
  },
  ...
]
```

- 2c.（8%）请你观察`venue`和`venueid`属性字段和 ICLR 2026 Conference OpenReview 主页显示的论文相关信息，理解它们之间的关联。进一步地，请基于你理解得到的先验信息，尝试编写代码以整理出这 1000 篇论文中，分别隶属于 Accept (Oral) / Accept (Poster) / Conditional Accept (Oral) / Conditional Accept (Poster) / Reject / Withdrawn Submission / Desk Rejected Submission`的论文数量各自有多少篇，最后请检查它们的数量总和是否正确。

3.（42%）OpenReview 的最大特色在于其公开的评审机制，这一点尤其体现在 ICLR 会议的审稿过程中。在实际的数据仓库构建中，论文的“基本信息”与“评审意见”往往分别存储在不同的数据表中。本任务使用论文的编号`number`属性，定向获取评审数据，清洗出核心的数值指标。

- 3a.（10%）请对`paper_info_cleaned.json`文件中前 100 篇论文提取审稿意见（Official Review）。



Official Review of Submission25369 by Reviewer aTGr

Official Review by Reviewer aTGr 03 Nov 2025, 14:12 (modified: 12 Nov 2025, 18:30) Everyone Revisions

Summary:
The paper introduces Common Corpus, a ~2T-token multilingual dataset for LLM pre-training that contains only data that are in the public domain or released under permissive open licenses. The authors document data sources, licensing, cleaning tools (Segmentext, OCRonos, Celadon, PII filtering), and present benchmark results for small Llama-style models trained solely on this corpus.

Overall, this is a valuable contribution to open LLM research. The motivation is strong, and the authors make extensive efforts to demonstrate the usefulness of the dataset for pre-training purposes.

However, several aspects including the very Euro-centric nature of the multilingual coverage and lack of quantitative information about ambiguous licenses need further clarification.

Soundness: 3: good
Presentation: 2: fair
Contribution: 3: good
Strengths:
1. Timely contribution to open-data infrastructure for LLMs, especially given that many existing datasets had to be taken down due to license issues.
2. Transparent and detailed documentation of dataset provenance, content, and filtering pipeline
3. Releases and discusses useful tools for OCR correction and sentence segmentation
4. Demonstrates that open data sourced from only permissive licenses can still yield competitive model performance.

Weaknesses:
1. The claim of "multilingual diversity" is overstated as the dataset is heavily Euro-centric. The top ten languages are all European, and there is little inclusion of African or Asian languages.
2. It is not clear how trustworthy the licensing information is. The authors do not report how many documents had ambiguous, conflicting, or missing license information.
3. For certain types of data, licenses can apply inconsistently across sub-components. It is not clear how these were disambiguated.
4. The data pre-processing pipeline, including Segmentext and OCRonos are underdescribed. It is not clear if any de-duplication was done or required.
5. The evaluation lacks detail on per-language performance and broader benchmark coverage.

Questions:
1. Can the authors provide a full language inventory and token count distribution, including low-resource languages?
2. How are ambiguous or missing licenses handled, and what proportion of the dataset do they represent?
3. What steps were taken to ensure deduplication across overlapping sources?

Flag For Ethics Review: No ethics review needed.
Rating: 8: accept, good paper (poster)
Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.
Code Of Conduct: Yes

Add: Public Comment

仿照 2a 的方式设置变量参数，将`params`中的`invitation`值修改为：

```
"invitation": "ICLR.cc/2026/Conference/Submission{number}/-/Official_Review"
```

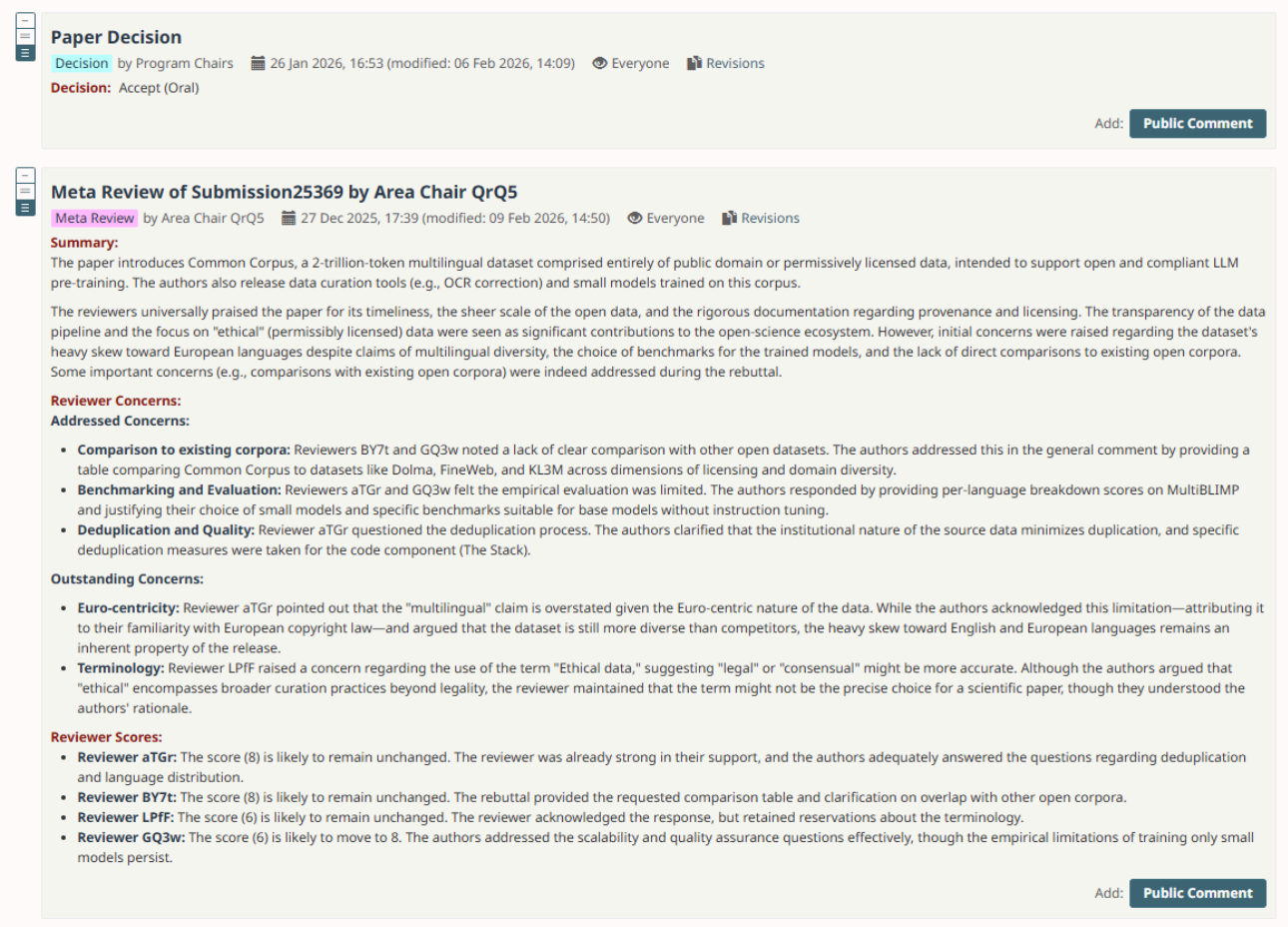
其中`{number}`表示论文编号。对`paper_info_cleaned.json`文件中每篇论文的信息字典，请你为每个 item 增加一个新的`reviews`字段（需提取`soundness`，`presentation`，`contribution`，`rating`和`confidence`），抽取存储相应值后将最终结果保存至`paper_info_cleaned_reviews.json`中。

- 最终所得数据的参考格式如下：

```
[
  {
    "id": "FPLNSx1jmL",
    ...
    "reviews": [
      {
        "soundness": 2,
        "presentation": 2,
        "contribution": 2,
        "rating": 2,
        "confidence": 4
      },
      ...
    ]
  },
  ...
]
```

  也就是说，与原文件`paper_info_cleaned.json`相比，新json文件中的每个item仅增加了`reviews`（以及五个子键值）的键值，其余属性键值保持不变。

- 3b.（16%）请继续对这 100 篇论文提取元评审意见（Meta Review）和最终决定（Decision）。



Paper Decision

Decision by Program Chairs 26 Jan 2026, 16:53 (modified: 06 Feb 2026, 14:09) Everyone Revisions

Decision: Accept (Oral)

Add: Public Comment

Meta Review of Submission25369 by Area Chair QrQ5

Meta Review by Area Chair QrQ5 27 Dec 2025, 17:39 (modified: 09 Feb 2026, 14:50) Everyone Revisions

Summary:

The paper introduces Common Corpus, a 2-trillion-token multilingual dataset comprised entirely of public domain or permissively licensed data, intended to support open and compliant LLM pre-training. The authors also release data curation tools (e.g., OCR correction) and small models trained on this corpus.

The reviewers universally praised the paper for its timeliness, the sheer scale of the open data, and the rigorous documentation regarding provenance and licensing. The transparency of the data pipeline and the focus on "ethical" (permissibly licensed) data were seen as significant contributions to the open-science ecosystem. However, initial concerns were raised regarding the dataset's heavy skew toward European languages despite claims of multilingual diversity, the choice of benchmarks for the trained models, and the lack of direct comparisons to existing open corpora. Some important concerns (e.g., comparisons with existing open corpora) were indeed addressed during the rebuttal.

Reviewer Concerns:

Addressed Concerns:

- **Comparison to existing corpora:** Reviewers BY7t and GQ3w noted a lack of clear comparison with other open datasets. The authors addressed this in the general comment by providing a table comparing Common Corpus to datasets like Dolma, FineWeb, and KL3M across dimensions of licensing and domain diversity.
- **Benchmarking and Evaluation:** Reviewers aTGr and GQ3w felt the empirical evaluation was limited. The authors responded by providing per-language breakdown scores on MultiBLIMP and justifying their choice of small models and specific benchmarks suitable for base models without instruction tuning.
- **Deduplication and Quality:** Reviewer aTGr questioned the deduplication process. The authors clarified that the institutional nature of the source data minimizes duplication, and specific deduplication measures were taken for the code component (The Stack).

Outstanding Concerns:

- **Euro-centricity:** Reviewer aTGr pointed out that the "multilingual" claim is overstated given the Euro-centric nature of the data. While the authors acknowledged this limitation—attributing it to their familiarity with European copyright law—and argued that the dataset is still more diverse than competitors, the heavy skew toward English and European languages remains an inherent property of the release.
- **Terminology:** Reviewer LPfF raised a concern regarding the use of the term "Ethical data," suggesting "legal" or "consensual" might be more accurate. Although the authors argued that "ethical" encompasses broader curation practices beyond legality, the reviewer maintained that the term might not be the precise choice for a scientific paper, though they understood the authors' rationale.

Reviewer Scores:

- **Reviewer aTGr:** The score (8) is likely to remain unchanged. The reviewer was already strong in their support, and the authors adequately answered the questions regarding deduplication and language distribution.
- **Reviewer BY7t:** The score (8) is likely to remain unchanged. The rebuttal provided the requested comparison table and clarification on overlap with other open corpora.
- **Reviewer LPfF:** The score (6) is likely to remain unchanged. The reviewer acknowledged the response, but retained reservations about the terminology.
- **Reviewer GQ3w:** The score (6) is likely to move to 8. The authors addressed the scalability and quality assurance questions effectively, though the empirical limitations of training only small models persist.

Add: Public Comment

该操作仍需设置变量参数，在提取Meta Review时，需将`params`中的`invitation`值修改为：

```
"invitation": "ICLR.cc/2026/Conference/Submission{number}/-/Meta_Review"
```

在提取Decision时，需将`params`中的`invitation`值修改为：

```
"invitation": "ICLR.cc/2026/Conference/Submission{number}/-/Decision"
```

对`paper_info_cleaned_reviews.json`文件中每篇论文的信息字典，请你为每个 item 增加一个新的`meta_review`字段（需提取`summary`，`reviewer_concerns`和`reviewer_scores`）和`decision`字段，抽取存储相应值后将最终结果保存至`paper_info_cleaned_final.json`中。

- 最终所得数据的参考格式如下：

```
[
  {
    ...
    "meta_review": {
      "summary": ...,
      "reviewer_concerns": ...,
      "reviewer_scores": ...
    },
    "desicion": "reject"
  },
  ...
]
```

也就是说，与原文件`paper_info_cleaned_reviews.json`相比，新json文件中的每个item仅增加了`meta_review`（以及三个子键值）和`decision`的键值，其余属性键值保持不变。

- 3c.（16%）基于最终提取的 100 篇论文数据文件`paper_info_cleaned_final.json`，开展以下探索：
  - 请统计所有被接收（不含conditional accept）的文章和被拒绝（不含withdrawn submission和desk reject）的论文数量分别有多少？
  - 审稿人评分`rating`的取值范围和各取值的含义分别是什么？置信度`confidence`呢？
  - 审稿人均分（可以对`rating`取平均值得到）大于5分，但最终被拒稿的论文数量有多少篇？
  - 审稿人均分（可以对`rating`取平均值得到）小于5分，但最终被接收的论文数量有多少篇？
  - 如果将前两小问的审稿人均分，替换为按照置信度`confidence`加权所得的均分，按相同方式进行评判，最终结果如何？

4.（选做）现在，你已经得到了一个包含 100 篇论文的结构化 JSON 格式数据文件。请你查阅资料，尝试利用`sqlite3`或`mysql`等库的相关方法，将其存储进数据库以增加其可读性，并展示前几行的数据可视化结果。在处理过程中，可以考虑以下因素并进行预处理，随后再进行数据导入。

- 关系型数据库可能不支持部分类型的键值数据（或呈现形式不及预期），对此可根据语义进行灵活处理（例如将`authors`列表数据用逗号连接为文本类型数据）。
- 部分含有嵌套信息的数据无法直接导入关系型数据库，可考虑适当进行合并（例如用`"reviewer_ratings": "2, 2, 2, 4"`取代各审稿人的评分数据）。
- 进一步地，可通过数据模式的主键外键关系构建多个数据库模式，并进行相应存储。

格式要求

1. 请按具体任务分步编写代码，存储于`.ipynb`格式文件中用于复现，必要时可增加注释。

2. 实验报告必须涵盖任务列表中的所有内容和相应结果，并请存储于`.md`格式文件中。

参考资料

以下资料可能会对你顺利完成实验有所帮助。

1. 使用 Conda 配置虚拟环境与管理安装包：点击这里

2. Conda 轻量级版本 Miniconda 的安装地址：点击这里

3. 在 VSCode 中使用 Jupyter Notebook 进行代码实现：点击这里

4. Python 官方教程（版本：3.12.12）：点击这里